Tuning Report Adult Income Classifier

1. Data & Split

Adult Census Income dataset (OpenML **adult**, v2): **48,842 rows**, target = income > $50K (**11,687 positive / 37,155 negative**, ~23.9% positive).

Set	Rows	Used for
Train	34,189	Hyperparameter search (Task 2), learning curves & sweeps (Task 3), fitting the calibrated model (Task 4)
Validation	7,326	Calibration comparison + threshold selection (Task 4) only
Test	7,327	Final evaluation (Task 5) touched exactly once

2. Hyperparameter Search (Task 2)

RandomizedSearchCV, 5 fold StratifiedKFold, **n_iter=60** per model, scored on F1.

Model	Best CV F1	Best Parameters
Logistic Regression	0.6711	solver=liblinear, penalty=l1, C=2.976
Random Forest	0.6762	n_estimators=300, min_samples_leaf=2, max_features=sqrt, max_depth=None
Gradient Boosting (selected)	**0.7061**	**max_iter=200, max_depth=5, learning_rate=0.2, l2_regularization=0.1**

3. Bias / Variance Diagnosis (Task 3)

Logistic Regression (C sweep): best val F1 = 0.6711 at C≈46.4; train/val gap only 0.0053, barely widens (0.0058) even at the loosest C tested. High bias, low variance near its ceiling; more regularization or data won't help.

Random Forest (max_depth sweep): unrestricted depth → train F1 0.9999 vs val F1 0.6673 (gap 0.333, textbook overfitting). Even at the best val depth (20), gap is still 0.180. **Fix:** cap max_depth well below 20 or raise min_samples_leaf; more training data would also help since variance, not bias, is the bottleneck.

4. Calibration & Threshold Selection (Task 4)

Model	Brier Score (validation, uncalibrated)
Gradient Boosting	**0.0856**
Random Forest	0.0922
Logistic Regression	0.0960

Calibrating Gradient Boosting (sigmoid, 5 fold) improved this further: **0.0856 → 0.0835**.

Threshold sweep on validation selected **best_threshold = 0.40** (peak F1 = 0.737):

Threshold	Precision	Recall	F1
0.10	0.501	0.945	0.654
0.25	0.618	0.861	0.720
0.35	0.689	0.788	0.735
0.40 (selected)	**0.735**	**0.739**	**0.737**
0.50 (default)	0.804	0.669	0.730
0.65	0.888	0.518	0.654

5. Final Test Set Evaluation (Task 5)

Final pipeline: **Gradient Boosting, calibrated (sigmoid), threshold 0.40**, evaluated once on the untouched 7,327 row test set:

Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
0.8657	0.7128	0.7347	0.7236	0.9272	0.8291

Test F1 (0.724) tracks the validation tuned figure (0.737) closely no meaningful drop from choosing the threshold on a different split.

Confusion matrix (test set)	Tuned (0.40)	Default (0.50)
True Neg / False Pos	5055 / 519	5262 / 312
False Neg / True Pos	465 / 1288	623 / 1130

Moving from 0.50 to 0.40 costs 207 more false positives but recovers 158 false negatives a deliberate trade toward recall, matching the F1 objective.

6. Artifact & Production Notes

- Saved as adult_income_pipeline.joblib: {pipeline, threshold, model_name, best_params, sklearn_version, random_state} the tuned 0.40 threshold ships with the model.
- The saved Pipeline includes feature engineering, imputation, scaling, and one hot encoding no separate preprocessing needed at inference.
- Re validate calibration and re run the threshold sweep if retrained on new production data, since both are sensitive to shifts in class balance and feature distributions.